Apparent polar wander paths and the close of late Archean crustal transpression, northern Ontario

G. J. Borradaile, T. J. Lemmetty, and T. Werner[1]

Geology and Physics Department, Lakehead University, Thunder Bay, Ontario, Canada

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[1] Lamprophyre dikes of the southern Superior Province of the Canadian Shield crosscut Archean structures, including a late Archean unconformity and its overlying younger Archean Timiskaming sedimentary and volcanic rocks. The dikes also cut the single schistosity that formed in these rocks. Nevertheless, the lamprophyre dikes show a weak internal tectonic fabric that is approximately parallel to the schistosity in the bedrock and oblique to dike walls. The consistent orientation of this internal tectonic fabric has been recognized in every lamprophyre dike using anisotropy of magnetic susceptibility (AMS). Thus the lamprophyre dikes provide a useful chronological marker, trapping the last pulse of Archean tectonism. Unfortunately, our previous attempts to compare the lamprophyre paleopoles with those of the age-calibrated apparent polar wander path (APWP) failed because it was not possible to isolate separate young and old components of magnetization. That failure occurred due to the overlap of blocking temperatures or coercivities for the different characteristic vector components [*Borradaile et al.*, 1994a]. Using new cores from the same specimens, here we distinguish successfully between consistently oriented vector components by using low-temperature demagnetization (LTD) before thermal demagnetization. Although the paleopoles appear well defined, comparisons with published APWPs for the Superior Province from late Archean [*Irving*, 1979; *Dunlop*, 1983; *Symons et al.*, 1994] to early Proterozoic [*Buchan et al.*, 2000] are difficult. Moreover, the earlier parts of the paths are progressively less precise and complete. Nevertheless, A component remanences for the dikes give a paleopole near the path at ~2650 Ma. For a Timiskaming volcanic breccia with a U-Pb age of 2696 Ma the A component paleopole may lie between 2600 and 2650 Ma. B component remanences yield paleopoles in the NW Pacific region, incompatible with any proposed Archean path but perhaps representing some much younger Proterozoic remagnetization event (e.g., Keweenawan, ~1100 Ma). *INDEX TERMS:* 1518 Geomagnetism and Paleomagnetism: Magnetic fabrics and anisotropy; 1525 Geomagnetism and Paleomagnetism: Paleomagnetism applied to tectonics (regional, global); 1527 Geomagnetism and Paleomagnetism: Paleomagnetism applied to geologic processes; 1533 Geomagnetism and Paleomagnetism: Remagnetization; 8157 Tectonophysics: Plate motions—past (3040); *KEYWORDS:* apparent polar wander path, paleomagnetism, lamprophyre, Archean, Proterozoic, magnetic fabrics, AMS

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1. Introduction

[2] Tectonic studies of the Archean and Paleo-Proterozoic rocks of northwestern Ontario have benefited greatly from modern high-precision geochronology, mainly using U-Pb. In conjunction with paleomagnetic work, this may chronologically calibrate paleopoles along an apparent polar wander path (APWP) that provides a guide to the relative ages of other sites for which geochronology is lacking but where more readily obtained paleomagnetic data are available. Such relative dating, although of low precision, perhaps as poor as ±20 Ma even in the Phanerozoic, permits crude estimates of ages where other methods fail, e.g., in the case of mineralization and tectonic deformation.

[3] Four caveats require consideration. First, true polar wander must be negligible or absent. Whereas there is some debate even for Phanerozoic time, there is no reason to suppose that a geomagnetic dipole field has been substantially nonaxial for any meaningful period of time during earth history [*Butler*, 1992; *Merrill and McElhinny*, 1983; *van der Voo*, 1993].

[4] Second, the paleomagnetic sites should come from a terrane with considerable structural integrity. Any relative motion within the studied region, for example, on unde-

[1]Now at Institute of Geophysics, Polish Academy of Sciences, Warsaw, Poland.

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tected or poorly documented structural boundaries will invalidate the paleopole positions determined from the sites. This relates also to the correspondence that exists between the paleomagnetic data and any geochronological information. The blocking temperature of thermoremanent magnetization (TRM) or chemical remanent magnetization (CRM) and the closure temperature of a radiogenic isotope series cannot be readily reconciled. Although high precision geochronology now routinely achieves 2σ precision of ±2 Ma for ages of 2500 Ma to 3000 Ma, in this region [e.g., *Hanes et al.*, 1994; *Horn et al.*, 2000; *Krogh and Moser*, 1994], the ages associated with previously published paleopole locations were usually much less precise K-Ar ages. In general, both paleomagnetic signals and K-Ar radiometric ages may be reset by metamorphic and structural events. However, in the case of most metamorphic and plutonic rocks, one expects K-Ar ages to be set long before the characteristic paleomagnetic vector is blocked in.

[5] Third, at the site level, the rocks should not have been substantially strained, or even stressed with negligible strain, if the preexisting paleomagnetic vectors are to be used to determine paleopole positions. For metamorphic rocks, the complications go beyond the tilt corrections which successfully restore the orientations of paleomagnetic vectors in rigidly rotated units that lack penetrative deformation [*MacDonald*, 1980]. Field studies show that finite strain deflects remanence vectors in metamorphic rocks [*Hirt et al.*, 1986; *Lowrie et al.*, 1986; *Stamatakos and Kodama*, 1991]. Under limited conditions involving low strain or minimum recrystallization, the sense and even the amount of the vector's deflection sometimes agree surprisingly well with simplified continuum models of rock deformation [*Borradaile*, 1993, 1997; *Kodama*, 1988]. However, despite progress in recognizing the effects of tectonism on paleomagnetic records, it is still unwise to attempt any restoration of naturally deformed paleomagnetic vectors from penetratively deformed rocks on a routine basis, especially where so much importance is attached to the locations of their paleopoles. To note just a few complications, it is known that different vector components rotate at different rates [*Borradaile*, 1992, 1997; *Jackson et al.*, 1993], depending on their coercivity, and that ephemeral stresses may be more effective than finite strain in controlling the orientation of the characteristic vector [*Borradaile*, 1994]. The latter may be considered as stress demagnetization or stress remagnetization where different coercivity components respond differently in controlled triaxial laboratory rock deformation experiments [*Borradaile*, 1992; *Jackson et al.*, 1993]. The effects of stress, rather than strain deflection of remanence, may be common during the closing stages of natural deformation. Field evidence has been debated for the phenomenon [*McWhinnie et al.*, 1990; *Hudson et al.*, 1989; *Werner and Borradaile*, 1996]. Although our lamprophyre dikes were affected by a feeble strain in the waning stage of regional compression, their magnetic records are postdeformation remagnetizations that do not require substantial restoration. Our new paleopole data and most of the paleopole data that we review do not use tectonic corrections. The southern Superior craton was stabilized between 2700 and 2600 Ma [e.g., *Wyman and Kerrich*, 1988; *Corfu and Stott*, 2000], after which further metamorphism, penetrative deformation and significant mineralization did not occur [*Williams et al.*, 1992]. However, in some earlier studies of Proterozoic rocks and structures, tilt corrections were necessary for postcratonization rigid body rotations, but these were due to localized faulting, mostly in the brittle regime.

[6] Fourth, a suitable APWP must be available as a calibration tool. Two APWPs have been proposed for the Superior Province of the Canadian Shield in northwestern Ontario. The most recent APWP combines very high quality paleomagnetic data and high precision U-Pb geochronology from postcratonization dikes [*Buchan et al.*, 1990, 1993, 1996, 2000] (Figure 1, track 2). The expected correspondence of magmatic and paleopole ages is close, probably <5 Ma. Dike emplacement requires brittle crustal conditions and where the dikes are not subsequently tilted or deformed, they satisfy all the requirements for high precision paleomagnetic work at numerous locations in the Canadian Shield [e.g., *Buchan et al.*, 1990, 1993, 1996; *Ernst and Buchan*, 1993; *Ernst et al.*, 2000; *Halls et al.*, 1994]. Brittle conditions require depths <2 km at geothermal gradients of $\sim 30^{\circ}\mathrm{C\ km^{-1}}$ so that in most cases the remanences acquired could be straightforward TRMs without complications from metamorphic alteration and associated chemical remagnetization.

[7] Unfortunately, as we will show, our dikes are late syntectonic and a different and necessarily less precise APWP is used. It was obtained from plutonic rocks and mostly uses low-precision K-Ar geochronology [*Constanzo-Alvarez and Dunlop*, 1988; *Constanzo-Alvarez et al.*, 1993; *Dunlop*, 1979, 1983, 1984a, 1984b, 1985; *Irving*, 1979; *Symons et al.*, 1994] (Figure 1a, track 1). Whereas the geochronological data are much less precise than for track 2, paleopole positions have been repeatedly confirmed along these tracks from thermochemical remanences acquired during postmetamorphic cooling at the close of Archean deformation in this region (2700–2600 Ma). Track 2 commences in the latest Archean, and if one considers a possible antipodal position of the paleopole due to a geomagnetic reversal (Figure 1, inverse track 2), it could be contiguous with track 1 between 2470 and 2500 Ma.

2. Sampling and Structural Chronology

[8] The Superior Province comprises several E-W terranes of alternating greenstone and metasedimentary character. The minor intrusions we studied crop out in the Wabigoon greenstone, Quetico metasedimentary, and Shebandowan greenstone terranes at the southern edge of the shield. Their metamorphism and cooling have been assigned various geochronological ages around 2720–2660 Ma [*Blackburn et al.*, 1992; *Williams*, 1992a, 1992b; *Wyman and Kerrich*, 1988], generally termed the Kenoran tectonothermal event. Whereas the formation of the volcanic sequences must predate this cooling event, the plutonic tonalitic gneiss plutons which form their basement yield U-Pb ages of 2750 Ma [*Corfu and Stott*, 2000]. The erosion of the Quetico, Wabigoon, and Shebandowan terranes, including the unroofing of the underlying tonalitic gneissic plutons, preceded the deposition of slightly younger Archean shallow water clastic sequences. These are

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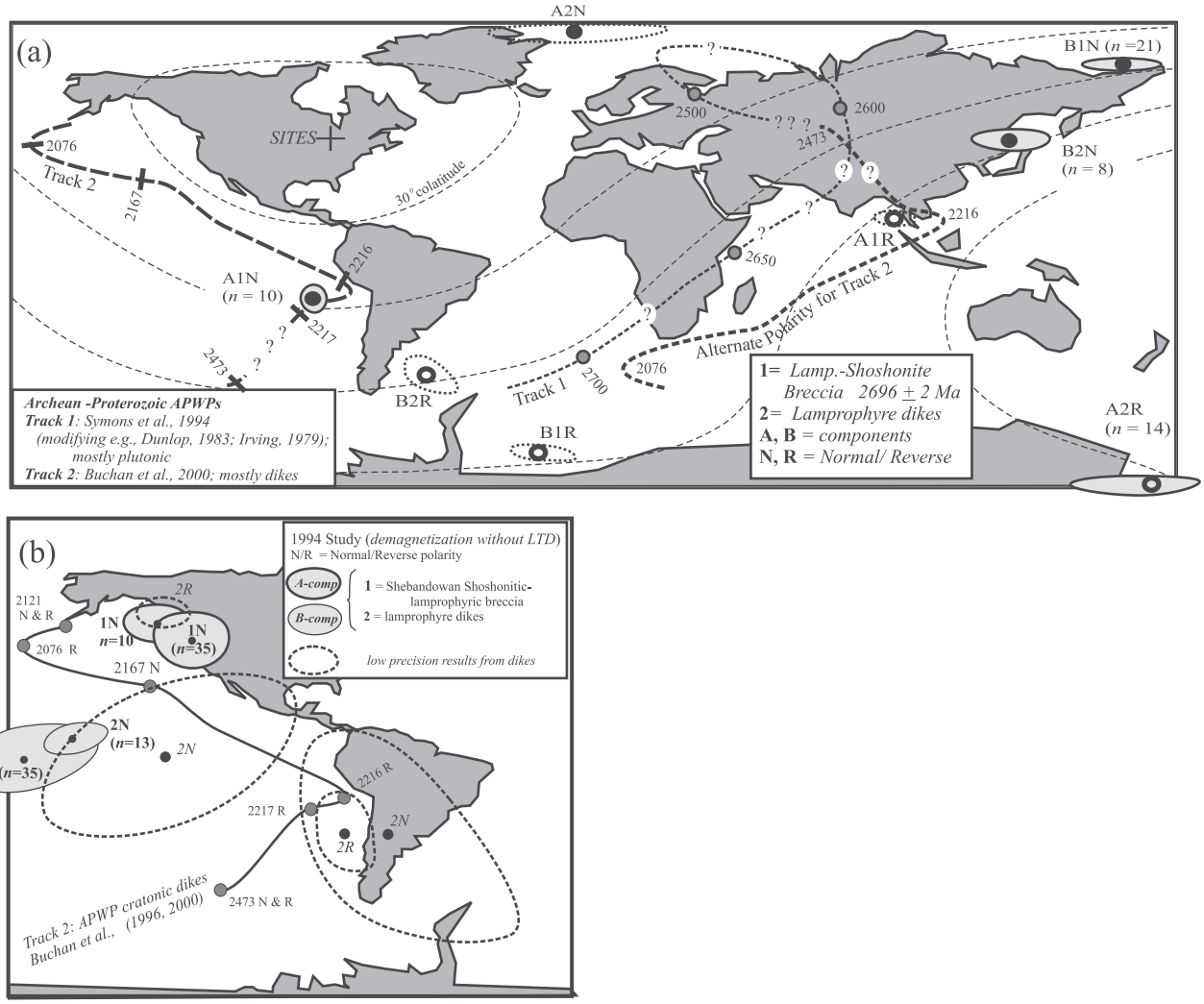


Figure 1. (a) Curves of colatitude at 30° intervals are shown about the mean site location in northern Ontario (272°E, 48°N). Paleopoles are shown with confidence cones calculated assuming Fisher statistics from this study. The A and B components were obtained by low-temperature demagnetization (LTD) followed by thermal demagnetization. The components for the lamprophyre dikes (A2N) and for the volcanic breccia (A1R) are associated with appropriate positions on track 1, at ~2550 and ~2625 Ma. (b) Different and poor results were obtained from the same specimens in an earlier study which did not use LTD before thermal or AF demagnetization [*Borradaile et al.*, 1994a].

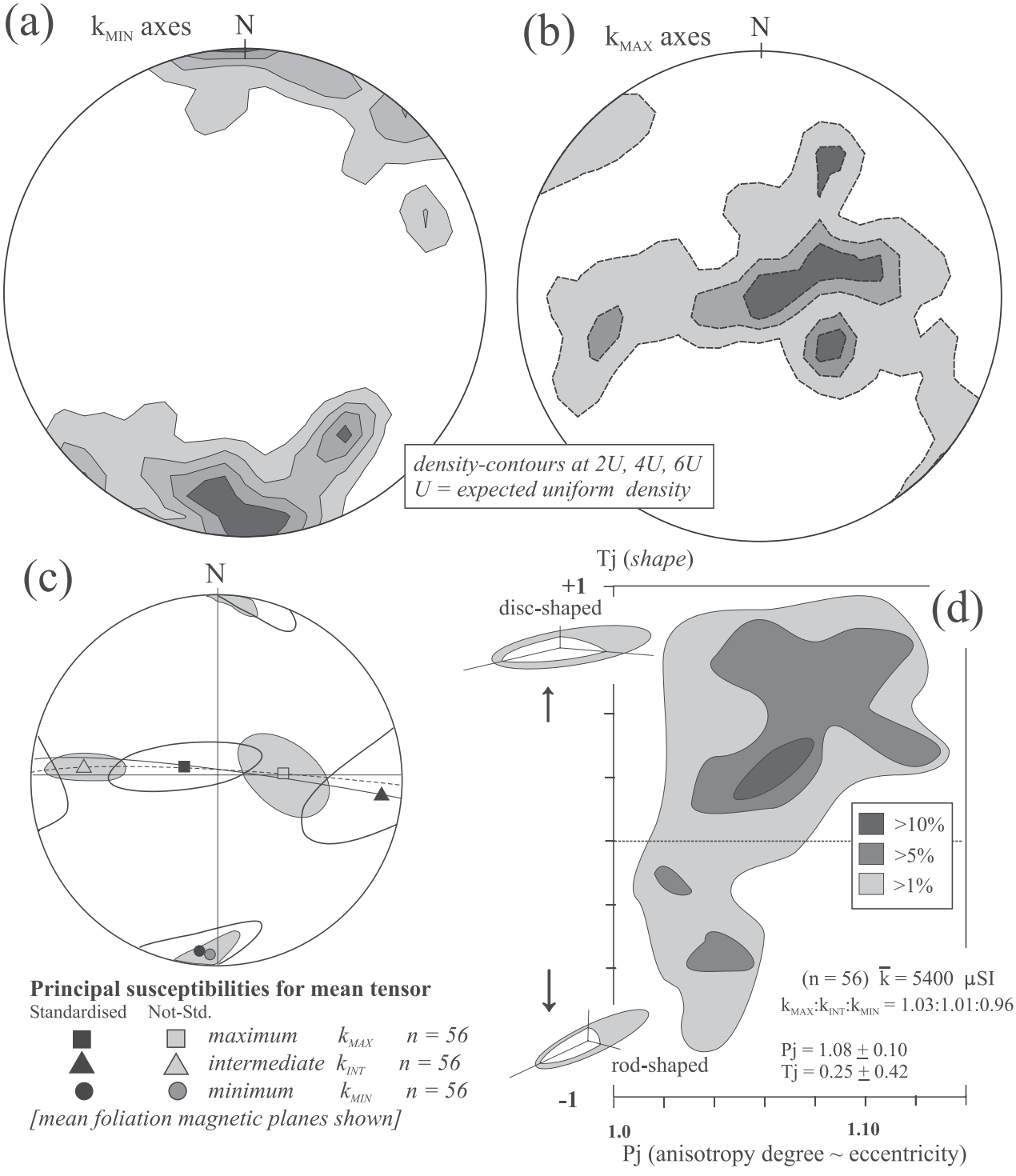
informally grouped as the Timiskaming sedimentary group. Although an angular unconformity and a metamorphic break separate the Timiskaming sequences from the older Archean rocks which underlie them, they share a similarly oriented schistosity and similarly oriented principal strain directions. Timiskaming sedimentation occurred in fault-bounded outliers either during a transient period of crustal tension, after which NNW-SSE compression soon resumed, or perhaps during continuous dextral transpression that permitted deposition in pull-apart basins. Subsequently, some short period of at least localized crustal tension was required for the intrusion of posttectonic lamprophyre dikes but NNW-SSE compression soon resumed, imposing a foliation in the dikes, subparallel to the cleavage in the Timiskaming and older Archean country rocks. Compression appears to have been accompanied by dextral shear as inferred from sequentially developed microfabrics, which indicate shear sense [*Borradaile and Dehls*, 1993; *Borradaile et al.*, 1993, 1994b; *Werner and Borradaile*, 1996].

[9] The lamprophyre dikes are narrow (<2 m), nonplanar, discontinuous, and steeply dipping sheets with a fine-grained to medium-grained, mafic groundmass bearing mica and amphibole phenocrysts, typical of the lamprophyre facies [*Mitchell*, 1994]. These have unstable orientations, alternately following schistosity or slaty cleavage in the country rock, or bedding, or fractures. The dikes postdate the Timiskaming sediments and its schistosity and they are found in the Quetico, Wabigoon, and Shebandowan subprovinces in the present study but also more widely in the Southern Superior Province [*Wyman and Kerrich*, 1988]. Near Shebandowan, dikes may be seen transecting the unconformity at the base of the Timiskaming sequence, following the schistosity in the Timiskaming rocks or slightly folded within the Timiskaming rocks. Thus the

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lamprophyre swarm overlaps the last pulse of Archean deformation [*Borradaile and Brown*, 1987]. Whereas dike intrusion requires brittle conditions, indicating cratonization, many dikes also show a subtle internal alignment of mafic minerals that is consistently inclined to dike margins [e.g., *Borradaile and Brown*, 1987] and parallel to the regional metamorphic foliation. We interpret this, as in other regions [*Borradaile*, 1986], to the persisting ductility of the late syntectonic lamprophyre. Similar late reactivation of the NNW-SSE regional compression in northern Ontario has also been detected in cryptic magnetic fabrics in igneous granite plutons that are considered posttectonic and post-metamorphic by all traditional field criteria [*Borradaile and Kehlenbeck*, 1996].

[10] At Shebandowan, a shoshonitic volcanic breccia also crops out in the Timiskaming outlier. Although field relations between the breccia and the sediments are unclear, these rocks are also considered to represent a late tectonic magmatic association [*Morrison*, 1980; *Wyman and Kerrich*, 1988]. They provide a precise U-Pb age of 2696 Ma [*Corfu*



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and Stott, 1986] and provide a group of our paleomagnetic subsets.

3. Petrofabrics and Magnetic Fabrics

[11] The tectonic foliation in the lamprophyres is poorly developed and difficult to measure in the field. However, it is now well known that the orientation distribution of minerals in a rock is expressed faithfully by the anisotropy of magnetic susceptibility (AMS), which blends the bulk susceptibility and anisotropy contributions from all the minerals [*Borradaile and Henry*, 1997; *Hrouda*, 1982; *Jackson and Tauxe*, 1991; *Tarling and Hrouda*, 1993]. Ferromagnesian silicates and multidomain (MD) magnetite dominate the lamprophyre bulk susceptibility. However, the anisotropy of susceptibility mainly reflects the preferred crystallographic alignment of silicates, which, despite lower bulk susceptibility, are more anisotropic than magnetite.

[12] The dike AMS defines a consistently oriented fabric parallel to the regional schistosity and AMS fabric of the country rock. Minimum susceptibility axes define the normals to magnetic foliation and these k_{MIN} axes are reasonably unimodally distributed, as are the k_{MAX} axes (Figures 2a and 2b). It is therefore meaningful to characterize them using the mean tensor and the 95% confidence cones for its axes (Figure 2c). The calculation may be performed using the AMS specimen tensors, directly as determined, or they may be normalized by dividing each specimen's principal axes by the specimen's bulk susceptibility [*Jelinek*, 1978]. In this way, standardization of the specimen tensors to unit value bulk susceptibility suppresses the bias in the mean tensor caused by specimens with anomalously high bulk susceptibility and unusual orientation. Here, the standardized mean tensor has less well defined principal axes and, moreover, the confidence cones do not have orthorhombic symmetry (Figure 2c). This usually suggests the presence of noncoaxial subfabrics [*Borradaile*, 2001]. In this context, the subfabrics may be the dike magma flow fabric and its tectonic overprint, respectively. Magnetic fabrics show that the tectonic fabric dominates, with an E-W vertical foliation and steep east plunging mineral lineation. The tectonic overprint is congruent with the metamorphic foliation and strain markers in the country rocks [*Borradaile et al.*, 1993; *Werner and Borradaile*, 1996].

[13] The mean tensor summarizes the regional orientation distribution of specimens' principal axes. However, the shape of the orientation distribution ellipsoid defines a regional sense of fabric symmetry, in the prolate-oblate range [e.g., *Woodcock*, 1977]. This may be inferred from the shapes of the confidence cones. For example, elongate, coplanar confidence cones about the mean tensor's maximum and intermediate axes define a disk-shaped fabric ellipsoid, tending toward the oblate end-member. For our lamprophyres, a regional L-S mineral orientation distribution fabric is inferred, notwithstanding the difficulties of isolating the contributing subfabrics.

[14] Shapes of individual specimen tensors reveal a different sense of fabric symmetry. This need not necessarily agree with the ellipsoidal shape of the orientation distribution and it is less useful in interpreting regional kinematics. Primarily, these reflect the degree of mixing of anisotropies and of bulk susceptibilities for different minerals within the specimens [*Borradaile and Henry*, 1997]. Individual specimen anisotropy is revealed by the plot of Figure 2d, using *Jelinek*'s [1981] parameters that separate shape ($+1$ [oblate] $\geq Tj \geq -1$ [prolate]) from the degree of anisotropy (Pj), which expresses the eccentricity of the magnitude ellipsoid. Pj ranges upward from unity for the isotropic case. Shape and anisotropy degree therefore lie on distinct Cartesian axes, an advantage over the Flinn plot of structural geology. Most individual specimens have disk-shaped fabric ellipsoids ($0 \leq Tj \leq +1$) close to the neutral shape ($Tj \sim 0$) due to the mixing oblate mineral AMS of mafic silicates with MD magnetite.

4. Paleomagnetic Procedure

[15] Previous paleomagnetic work on these lamprophyres identified few stable remanences, with anomalous paleopoles or scattered vectors [*Borradaile et al.*, 1994a] (Table 1 and Figure 1b). At most sites only one component was identified; directions were inconsistent from site to site, and at only two sites were two components isolated. Neither thermal nor alternating field demagnetization was successful. Vector plots were typically unstable. Vector plots did not show clear turning points and graphs showing intensity decrease rarely showed distinct unblocking of A and B components (Figures 3a and 3c). The few stable primary vectors were scattered (Figure 4a) and yield uninterpretable paleopoles (Figure 1b).

[16] This unsatisfactory and common situation was attributed to the overlap of blocking temperatures or coercivities of different vector components during experimental demagnetization [*Hoffman and Day*, 1978]. Grains that respond similarly either to thermal demagnetization (TD) and alternating field demagnetization (AF) may share differently oriented magnetic components. As one step of incremental demagnetization removes part of one directional compo-

Figure 2. (opposite) Lamprophyre dikes are variably oriented and rarely planar but show a consistently oriented internal foliation, inclined to dike margins. This internal fabric is consistently oriented through a large part of northwestern Ontario and parallel to the regional metamorphic foliation in the country rocks. The internal fabric is consistently recorded by anisotropy of magnetic susceptibility (AMS), which blends the AMS contributions of all minerals into a composite orientation distribution. (a) AMS reveals an E-W foliation of aligned minerals. (b) AMS shows steep east plunging mineral lineation. (c) Since the density contours reveal a unimodal regional orientation distribution for secondary fabric in the dikes, it is reasonable to determine the mean tensor's axes and their confidence [*Jelinek*, 1978]. The contribution of anomalously oriented, high-susceptibility specimens to the sample is suppressed where specimen tensors are normalized to unit value bulk susceptibility. Shapes of the confidence regions indicate that the orientation distribution ellipsoid for the sample suite is of the L-S shape. (d) The shapes of tensor ellipsoids for individual specimens shown on a Jelinek plot. Tj represents shape ($+1$ [oblate] $\geq Tj \geq -1$ [prolate]). The aspect ratios of the magnitude ellipsoid are summarized as the degree of anisotropy ($1 \leq Pj \leq \infty$; 1 = sphere).

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Table 1. Previous Paleomagnetic Data Summarized From *Borradaile et al.* [1994a] Thermal and AF Demagnetization Without Low-Temperature Demagnetization[a]

		Fisher Statistics				Paleopole	
Site[b]	*n*	D, deg	I, deg	α_{95}	k	Latitude	Longitude
[A] Shebandowan shoshonite volcanic breccia	35	277	77	15	42	45.1	233.8
[B] Shebandowan shoshonite volcanic breccia	35	267	5	38	1	−0.2	183.9
[A] Shebandowan shoshonitic lamprophyre dikes	10	294	74	23	5	51.2	223.4
[B] Shebandowan shoshonitic lamprophyre dikes	13	261	34	17	7	7.9	199
Four-way School lamprophyre dikes	7	339	−23	71	1.6		
Duckworth Road lamprophyre dikes	11	351	29	26	4		
Dawson Road shoshonitic volcanic breccia	15	54	51	80	1.2		
Dawson Road lamprophyre dikes	6	306	75	18	14		

[a]These specific localities are near the City of Thunder Bay; locations in which other lamprophyre dikes may be found are listed in Table 2. See Figure 1b. D, declination; I, inclination.
[b][A], [B] indicate where components were distinguishable.

nent, it unfortunately also removes part of a differently oriented component. Consequently, the ideal isolation of one directional component from another in the chronological order of their acquisition is not achieved. Although it is not yet routine in paleomagnetic applications, a long-known and recently studied demagnetization phenomenon called low- temperature demagnetization (LTD) sometimes rectifies this problem [*Dunlop and Özdemir*, 1997]. Magnetically shielded specimens are cooled to liquid Nitrogen temperature (77 K) and then permitted to warm back to room temperature inside the shield. As the specimens warm through two important transition temperatures, the isotropic point (∼114 K) and the Verwey temperature (∼120 K), domain walls mobilize in polydomain magnetite, and their associated magnetic moment is randomized in zero field. Reasonable theoretical and experimental justification, as well as some successful applications [e.g., *Dunlop and Argyle*, 1991; *King and Williams*, 2000; *Muxworthy and McClelland*, 2000] encouraged us to apply LTD prior to thermal demagnetization studies. We have demonstrated its effectiveness prior to thermal or AF demagnetization in a companion paper [*Borradaile et al.*, 2003]. That study used a variety of magnetite-bearing igneous rocks to which multicomponent remanences were experimentally applied in the laboratory. The remanences imposed were isothermal and anhysteretic remanent magnetizations (IRMs and ARMs) and the magnetite in those experiments may have lower dislocation densities because the materials were not metamorphosed or tectonically deformed. Nevertheless, it encouraged us to apply LTD here, before we applied thermal demagnetization. We prepared new drill core from the same oriented hand specimens that we used in the earlier study. Three cycles of LTD achieved optimum demagnetization and removed 20–50% of the remanence for MD magnetite. Subsequent incremental thermal or AF demagnetization then yielded distinct turning points between successively acquired vector components. Two examples compare the results obtained in the 1993 campaign without LTD, with those here using LTD before thermal demagnetization (Figure 3). Using only thermal or AF demagnetization, the vector plots did not always show clear turning points and the vector segments were rather unstable in orientation (Figures 3a and 3c). These examples were selected because the A and B component remanences were almost antiparallel. Thus the intensity plots, shown as graphs of normalized remanence, indicate the degree to which unblocking discriminates between the components. In the case of the earlier study, without LTD, thermal demagnetization (Figure 3a) barely distinguished the unblocking temperature of the A component and AF demagnetization failed completely (Figure 3c). In the present study, when cores from the same hand specimens were exposed to three cycles of LTD, and then to thermal demagnetization, vector plots gave more distinct turning points and intensity graphs identified a distinct unblocking temperature for the A component (Figures 3b and 3d).

[17] In this study we applied three steps of low-temperature demagnetization (LTD1, etc., in Figure 3) before incremental thermal demagnetization at temperatures of 200, 300, 350, 400, 450, 475, 500, 550, and 580°C using a thermal demagnetizer (MMTD80 by Magnetics Measurements Corporation). It should be noted that the earlier study used a thermal demagnetizer of our own manufacture. Although it was inefficient in the number of specimens that it could process and lacked automated circuitry; its temperature control, temperature gradients and magnetic shielding were not inferior to the MMTD80 used in this study.

[18] Remanence was measured initially, and after each step of demagnetization, using a Molspin spinner magnetometer. The pretreatment by LTD permitted us to isolate straight vector components, separated by a sharp turning point. The B component is isolated after LTD treatment, by thermal demagnetization up to 400°C. Thereafter, the A component is unblocked at 450°C or 475°C and is stable until the remanence intensity is negligible, at ∼580°C, approximately the Curie point for magnetite. The magnetic properties and the paleomagnetic records show a geographically uniform distribution and thus they are grouped for the entire study, according to similarity in direction of the A components and of the B components. For the purposes of identifying vector components for PCA analysis, we find that three-dimensional, interactive rotatable vector plots are superior on the computer screen (e.g., Figures 4b and 4c). One can readily verify vectors whose stability may be concealed on a traditional vector plot by an unfortunate orientation with respect to a fixed projection plane.

5. Paleomagnetic Results

[19] The shoshonitic-lamprophyre breccia near Shebandowan is discussed first since these rocks have provided a precise U-Pb age of 2696 Ma for their magmatic crystalli-

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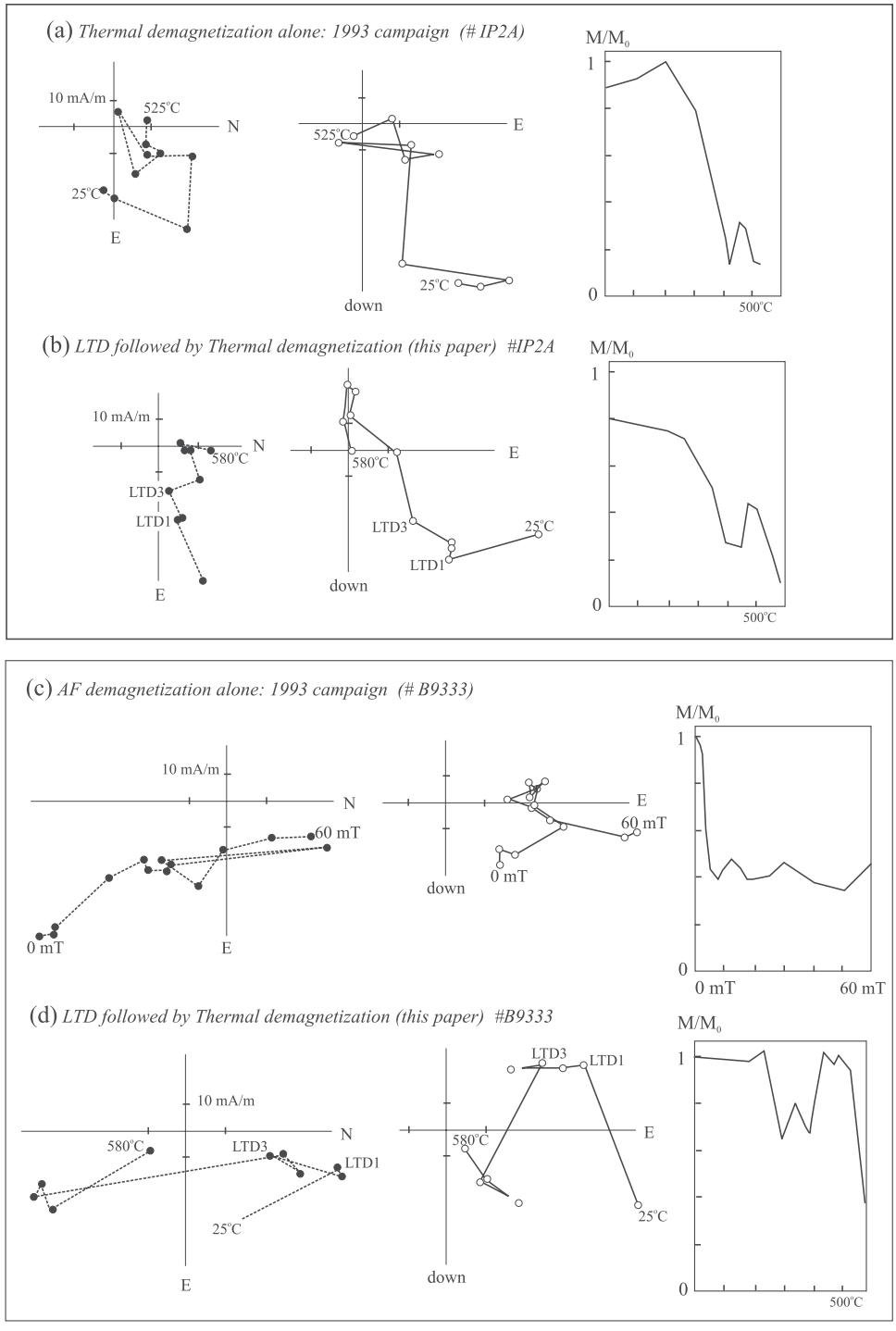


Figure 3. Demagnetization plots shown for cores from two hand specimens. (a, c) Results of an earlier study [*Borradaile et al.*, 1994b] which did not use LTD. (b, d) In this study, three cycles of LTD preceded thermal demagnetization, separating vector components more effectively. Rotated three-dimensional views show these vector components in Figures 4b and 4c.

zation [*Corfu and Stott*, 1986]. They are related petrogenetically to the lamprophyre dikes and form an extrusive or a subsurface intrusion. After the removal of spurious components by LTD two stable vector components are realized: a B component stable to about 450°C and an A component above that (Figures 5a and 5c and Table 2). The breccia appears to lack any evidence of tectonic deformation in the field, due to its massive nature and high competency. However, petrography shows evidence of metamorphism. Therefore its A1 components unblocking above 450°C are probably a CRM-TRM produced by an upward directed field during metamorphic, post deformation cooling (Figure 5a). The superimposed B1 components are due to a refractory CRM and are also quite well clustered but

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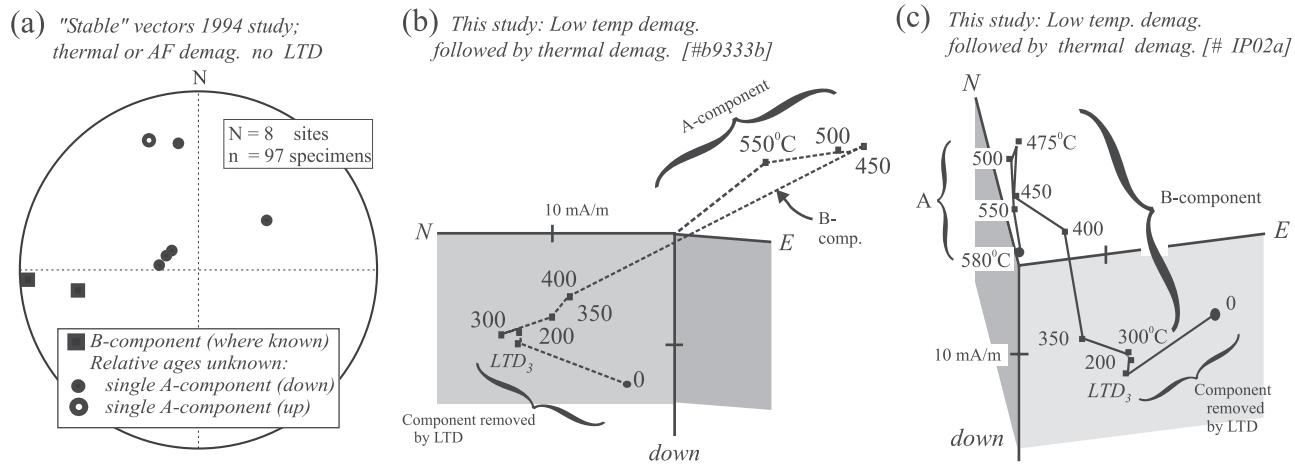


Figure 4. (a) An earlier study of these rocks yielded poorly defined and inconsistently oriented vector components [*Borradaile et al.*, 1994b]. (b, c) Stable vector components were identified from the same hand specimens in this study, by using low temperature demagnetization before thermal demagnetization. Three-dimensional plots rotated on a computer monitor reveal the components more readily than conventional two-dimensional vector plots (compare Figure 3). Vector components were defined by principal component analysis using four or more points [*Kirschvink*, 1980].

produced by a downward seeking field. Both components are present in the matrix and in the fragments. Both A and B component characteristic vectors have a reasonable unimodal distribution so that Fisher statistics have been used to determine paleopoles. Bingham statistics and the appropriate confidence region and symmetry plane have also been determined; the similar results confirm that the Fisher model is appropriate.

[20] The lamprophyre dikes also show two components (A2, B2), and although the B component is scarcer (Table 2), it is consistently oriented. The components' orientation distributions are suitable for the Fisher model, their statistics differing little from the Bingham model (Figures 5b and 5d). The A component is directed upward to the south whereas the B component is gently inclined NNW. For the purposes of interpretation and paleopole referencing in the next section and in Table 2, the components are distinguished as A1, B1 for the breccia and A2, B2 for the dikes. In all cases, the 95% confidence regions around the mean vector are <12° which is satisfactorily small for routine paleomagnetic work.

6. Interpretation

[21] Paleomagnetic data for the Superior Province of northwestern Ontario derives from two main sources which yield different APWPs (tracks 1 and 2, Figure 1a). One must make allowance for the longitudinal distortion at high latitudes with the map projection of Figure 1. However, we prefer a Cartesian map projection to a hemispherical one on which the extent of neither APWP would be entirely visible. Track 1 shows paleopoles mostly from plutonic and crystalline Archean metamorphic and igneous rocks that bear postmetamorphic chemical remagnetizations [*Dunlop*, 1979, 1983, 1984a, 1984b, 1985; *Irving*, 1979; *Irving and McGlynn*, 1981]. The initial apart of the APWP follows a broad path from the south Atlantic to central Asia but is poorly defined and covers much of the postmetamorphic cooling period of these late Archean rocks (2700–2500 Ma). Geochronological and paleomagnetic sites did not always correspond, and K-Ar techniques prevailed at the time most of the data was acquired. That lower precision geochronology is still useful for regional tectonic studies because K-Ar closure temperatures and paleomagnetic blocking temperatures both broadly respond to slow, plutonic, postmetamorphic cooling. The latter part of track 1 from late Archean and Paleo-Proterozoic rocks is much better defined by more recent studies, some of which used high-precision geochronology from Proterozoic rocks, or associated with Proterozoic events such as uplift of the Kapuskasing Structural Zone [e.g., *Constanzo-Alverez and Dunlop*, 1988; *Symons et al.*, 1994; *Symons and Vandall*, 1990; *Vandall and Symons*, 1990].

[22] Track 2 is more modern and is based mainly on postmetamorphic dikes from adjacent study areas (Figure 1a). It is particularly useful in precise paleogeographic reconstructions for the late Archean-Proterozoic transition 2400–2000 Ma [*Buchan et al.*, 1993, 1996, 2000, 2001]. Moreover, the paleomagnetic data correspond with sites where U/Pb geochronology fixed ages with 95% confidence limits of approximately ±2 Ma, providing a sound APWP for the Archean-Paleo-Proterozoic transition. However, field relations show that track 2 poles are defined by rocks that clearly postdate any late syntectonic compression, or reactivation of compression. They must therefore postdate the lamprophyre dikes and the Timiskaming volcanic agglomerate. Despite its poorer precision, track 1 may still be pertinent to us. It is tentatively suggested that track 1 may join track 2, when we consider the alternative polarity possibility for track 2 (Figure 1a). However, gaps and variable dating techniques render this suggestion highly tentative.

[23] We discuss the paleopoles derived from high temperature, "primary" A components first. "N" appended to the descriptor indicates Normal polarity, although the alternate polarity (R) should be considered, if appropriate.

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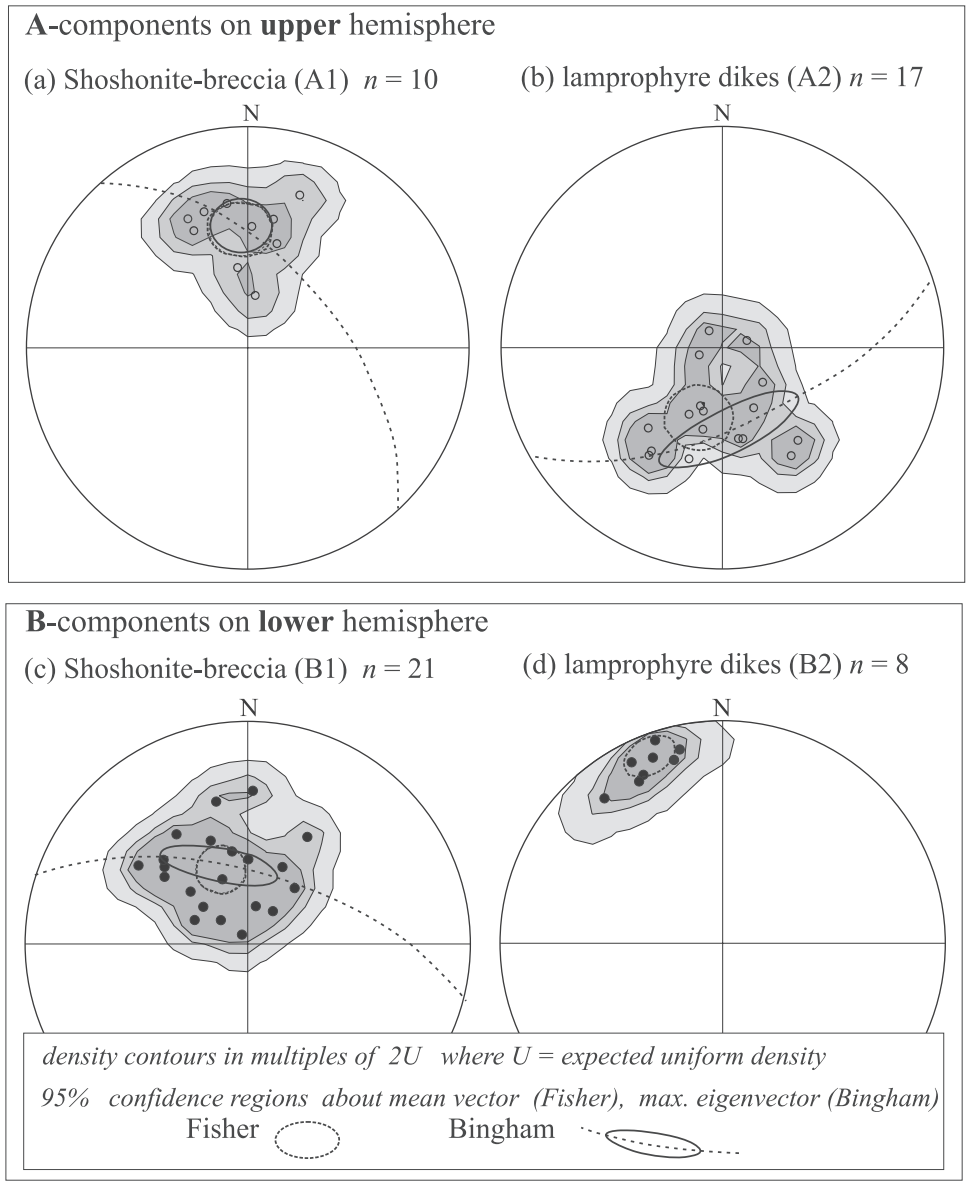


Figure 5. The A component vectors, unblocked above 450°C (a) for the volcanic breccia and (b) for the lamprophyre dikes. The B component vectors unblocked by the third stage of LTD and stable to 400°C or 425°C (c) for the breccia and (d) for the dikes. The distributions are sufficiently unimodal and circular in distribution on the sphere to use the Fisher model to determine confidence cones in all cases; the apparent elliptical distribution in Figure 5d is, of course, due to the projection. However, the Bingham confidence cones about the maximum eigenvectors and the symmetry plane for a possible Bingham distribution is shown for comparison.

Table 2. Paleopole Locations[a]

		Fisher Statistics				Paleopole Location			
	Description	*D*, deg	*I*, deg	α_{95}	*k*	Latitude	Longitude	*dm*	*dp*
A1 (*n* = 10)	shoshonitic lamprophyres (*n* = 3) breccia clasts (*n* = 3) breccia matrix (*n* = 4) 2696 ± 2 Ma [*Corfu and Stott*, 2000]	357.0	−45.2	11	21	−15.4	272.7	9.8	5.5
A2 (*n* = 17)	coarse-grained lamprophyres no geochronology	185.0	−66.1	12	12	86.6	352.7	7.9	6.0
B1 (*n* = 21)	shoshonite breccia (*n* = 3) matrix (*n* = 1) dikes (*n* = 17)	340.1	60.7	8.9	4	−74.5	343.3	6.6	4.4
B2 (*n* = 8)	dikes (*n* = 6) shoshonite-lamprophyre breccia (*n* = 2)	334.3	17.3	8.3	24	−45.2	307.8	8.2	4.1

[a]Mean location of sites is longitude 270°, latitude 48°N (see Figure 1a). Locations for shoshonitic breccia, several large outcrops with uncertain field relationships to one another between 48°32′N, 90°10′W and 48°32′N, 90°06′W. Lamprophyres dikes occur in numerous small outcrops between 48°42′N, 90°35′W and 48°37′N, 90°27′W and between 48°34′N, 90°03′W and 48°30′N, 89°36′W and near 48°34′N, 90°38′W.

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For the volcanic breccia, the paleopole (A1N) taken at face value lies fortuitously at ~2200 Ma on track 2. Field relations render this suggestion impossible but the alternate polarity (A1R) places it not far from track 1 at ~2650 Ma. That would be a satisfactory interpretation in view of the U-Pb age (2696 Ma) for primary recrystallization. The lamprophyre mean paleopole (A2N) lies at a high latitude in the Northern Hemisphere which, in view of the distortion in this map projection, is not far from the more precise part of track1 at ~2550 Ma. This interpretation is also reasonable in view of field relations. (The antipole corresponding to a reversed field, A2R, is disagreeably far from any relevant APWP.)

[24] The secondary, B components magnetizations are stable to at least 425°C, and therefore it is reasonable to consider them geologically significant CRMs or TRMs acquired during late metamorphic cooling. However, the pole positions taken at face value for the volcanic breccia (B1N) and for the dikes (B2N) do not correspond to any part of the Archean APWPs shown. Their antipodal equivalent positions that would have been generated in reverse fields (B1R, B2R) lie near the start of track 1 but their ages would be greater than that of the A components. The B1N and B2N paleopoles lie near the Proterozoic "Keewanawan" APWP in the northern Pacific region at ~1100 Ma, as discussed elsewhere [e.g., *Symons et al.*, 1994]. In that case, it may be difficult to propose an energy source that would trigger a widespread CRM or TVRM resulting in demagnetization temperatures as high as 450°C. However, dextral transpression was reactivated much later in northwestern Ontario, for example, causing Paleo-Proterozoic transcurrent fault motion, e.g., the Quetico Fault [*Williams*, 1992a, 1992b].

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References

Blackburn, C. E., G. W. Johns, J. Ayer, and D. W. Davis, The Wabigoon subprovince, in *The Geology of Ontario*, *Spec. Publ.4*(2), pp. 303–381, Ont. Geol. Surv., Sudbury, 1992.

Borradaile, G. J., The internal tectonic fabric of minor intrusions and their potential as regional paleostress indicators, *Geol. Mag.*, *123*, 665–671, 1986.

Borradaile, G. J., Experimental deformation of two-component IRM in magnetite-bearing limestone: A model for the behaviour of NRM during natural deformation, *Phys. Earth Planet. Inter.*, *70*, 64–77, 1992.

Borradaile, G. J., Strain and magnetic remanence, *J. Struct. Geol.*, *15*, 383–390, 1993.

Borradaile, G. J., Deformation and paleomagnetism, *Surv. Geophys.*, *18*, 405–435, 1997.

Borradaile, G. J., Magnetic fabrics and petrofabrics: Their orientation distributions and anisotropies, *J. Struct. Geol.*, *23*, 1581–1596, 2001.

Borradaile, G. J., and H. G. Brown, The Shebandowan group:"Timiskaming-like" Archean rocks in north-western Ontario, *Can. J. Earth Sci.*, *24*, 185–188, 1987.

Borradaile, G. J., and J. F. Dehls, Regional kinematics inferred from magnetic subfabrics in Archean rocks of Northern Ontario, Canada, *J. Struct. Geol.*, *15*, 887–894, 1993.

Borradaile, G. J., and B. Henry, Tectonic applications of magnetic susceptibility and its anisotropy, *Earth Sci. Rev.*, *42*, 49–93, 1997.

Borradaile, G. J., and M. M. Kehlenbeck, Possible crypto-tectonic magnetic fabrics in "posttectonic" granitoid plutons of the Canadian Shield, *Earth Planet. Sci. Lett.*, *137*, 119–127, 1996.

Borradaile, G. J., T. Werner, J. F. Dehls, and R. N. Spark, Archean regional transpression and Paleomagnetism in northwestern Ontario, Canada, *Tectonophysics*, *220*, 117–125, 1993.

Borradaile, G. J., M. M. Kehlenbeck, and T. W. Werner, A magnetotectonic study correlating late Archean deformation in north-western Ontario, *Can. J. Earth Sci.*, *31*, 1449–1460, 1994a.

Borradaile, G. J., R. A. Stewart, and T. Werner, Archean uplift of a subprovince boundary in the Canadian Shield, revealed by magnetic fabrics, *Tectonophysics*, *227*, 1–15, 1994b.

Borradaile, G. J., K. Lucas, and R. S. Middleton, Effectiveness of low-temperature demagnetization in the isolation of characteristic remanence, *Geophys. J. Int.*, in press, 2003.

Buchan, K. L., J. K. Mortensen, K. C. Card, and J. A. Percival, Paleomagnetism and U-Pb geochronology of diabase dike swarms of the Minto block, Superior Province, Canada, *Can. J. Earth Sci.*, *35*, 1054–1069, 1990.

Buchan, K. L., J. K. Mortensen, and K. D. Card, Northeast-trending Early Proterozoic dykes of southern Superior Province: Multiple episodes of emplacement recognized from integrated paleomagnetism and U-Pb geochronology, *Can. J. Earth Sci.*, *30*, 1286–1296, 1993.

Buchan, K. L., H. C. Halls, and J. K. Mortensen, Paleomagnetism, U-Pb geochronology, and geochemistry of Marathon dykes, Superior Province, and comparison with the Fort Francis swarm, *Can. J. Earth Sci.*, *33*, 1583–1595, 1996.

Buchan, K. L., S. Mertanen, R. G. Park, L. Pesonen, S.-A. Elming, N. Abrahamsen, and G. Bylund, Comparing the drift of Laurentia and Baltica in the Proterozoic: The importance of key paleomagnetic poles, *Tectonophysics*, *319*, 167–198, 2000.

Buchan, K. L., R. E. Ernst, M. A. Hamilton, S. Mertanen, L. Pesonen, and S.-A. Elming, Rodinia: The evidence from integrated paleomagnetism and U-Pb geochronology, *Precambrian Res.*, *110*, 9–32, 2001.

Butler, R. F., *Paleomagnetism: Magnetic Domains to Geologic Terranes*, 319 pp., Blackwell Sci., Malden, Mass., 1992.

Corfu, F., and G. M. Stott, U-Pb ages for late magmatism and regional deformation in the Shebandowan Belt, Superior Province, Canada, *Can. J. Earth Sci.*, *23*, 1075–1082, 1986.

Corfu, F., and G. M. Stott, Shenbandowan greestone belt, western Superior Province: U-Pb ages, tectonic implications, and correlations, *Geol. Soc. Am. Bull.*, *110*, 1467–1484, 2000.

Costanzo-Alverez, V., and D. J. Dunlop, Paleomagnetic evidence for post-2.55 Ga tectonic tilting and 1.1 Ga reactivation in the southern Kapuskasing Zone, Ontario, Canada, *J. Geophys. Res.*, *93*, 9126–9136, 1988.

Costanzo-Alverez, V., D. J. Dunlop, and L. Pesonen, Paleomagnetism of alkaline complexes and remagnetization in the Kapuskasing structural zone, Ontario, Canada, *J. Geophys. Res.*, *98*, 4063–4079, 1993.

Dunlop, D. J., A regional paleomagnetic study of Archean rocks from the Superior Geotraverse area, north-western Ontario, *Can. J. Earth Sci.*, *16*, 1909–1919, 1979.

Dunlop, D. J., Paleomagnetism of Archean rocks from north-western Ontario: Wabigoon gabbro, Wabigoon subprovince, *Can. J. Earth Sci.*, *20*, 1805–1817, 1983.

Dunlop, D. J., Paleomagnetism of Archean rocks from north-western Ontario: II. Shelley Lake granite, Quetico Subprovince, *Can. J. Earth Sci.*, *21*, 869–878, 1984a.

Dunlop, D. J., Paleomagnetism of Archean rocks from north-western Ontario: IV. Burchell Lake granite, Wawa Shebandowan Subprovince, *Can. J. Earth Sci.*, *21*, 1098–1104, 1984b.

Dunlop, D. J., Paleomagnetism of Archean rocks from north-western Ontario: V. Poohbah Lake alkaline complex, Quetico Subprovince, *Can. J. Earth Sci.*, *22*, 27–38, 1985.

Dunlop, D. J., and K. S. Argyle, Separating multidomain and single-domain-like remanences in pseudo-single-domain magnetites (215–540 nm) by low-temperature demagnetization, *J. Geophys. Res.*, *96*, 2007–2017, 1991.

Dunlop, D. J., and Ö. Özdemir, *Rock Magnetism: Fundamentals and Frontiers*, *Cambridge Stud. Magn.*, vol. 3, 573 pp., Cambridge Univ. Press, New York, 1997.

Ernst, R. E., and K. L. Buchan, Paleomagnetism of the Abitibi dike swarm, southern Superior Province, and its implications for the Logan Loop, *Can. J. Earth Sci.*, *30*, 1886–1897, 1993.

Ernst, R. E., K. L. Buchan, M. A. Hamilton, A. V. Okrugin, and M. D. Tomshin, Integrated paleomagnetism and U-Pb geochronology of mafic dikes of the eastern Anabar Shield region, Siberia: Implications for Mesoproterozoic paleolatitude of Siberia and comparison with Laurentia, *J. Geol.*, *108*, 381–401, 2000.

Halls, H., H. C. Palmer, M. P. Bates, and W. C. Phinney, Constraints on the nature of Kapuskasing Structural Zone from the study of Proterozoic dyke swarms, *Can. J. Earth Sci.*, *31*, 1182–1196, 1994.

Hanes, J. A., D. A. Archibald, M. Queen, and E. Farrar, Constraints from $^{40}Ar/^{39}Ar$ geochronology on the tectonothermal history of the Kapuskas-

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ing uplift in the Canadian Superior Province, *Can. J. Earth Sci.*, *31*, 1146–1171, 1994.
Hirt, A. M., W. Lowrie, and O. A. Pfiffner, A paleomagnetic study of tectonically deformed red beds of the Lower Glarus Nappe Complex, eastern Switzerland, *Tectonics*, *5*, 723–731, 1986.
Hoffman, K. A., and R. Day, Separation of multi- component NRM: A general method, *Earth Planet Sci Lett.*, *40*, 433–438, 1978.
Horn, I., R. L. Rudnick, and W. F. McDonough, Precise elemental and isotope ratio determination by simultaneous solution nebulization and laser ablation-ICP-MS: Application to U-Pb geochronology, *Chem. Geol.*, *167*, 405–425, 2000.
Hrouda, F., Magnetic anisotropy of rocks and its application in geology and geophysics, *Geophys. Surv.*, *5*, 37–82, 1982.
Hudson, M. R., R. L. Reynolds, and N. S. Fishman, Synfolding magnetization in the Jurassic Preuss Sandstone, Wyoming-Idaho-Utah thrust belt, *J. Geophys. Res.*, *94*, 13,681–13,705, 1989.
Irving, E., Paleopoles and paleolatitudes of North America and speculations about displaced terranes, *Can. J. Earth Sci.*, *16*, 669–694, 1979.
Irving, E., and J. C. McGlynn, On the coherence, rotation and paleolatitude of Laurentia in the Proterozoic, in *Precambrian Plate Tectonics*, edited by A. Kroner, pp. 561–598, Elsevier Sci., New York, 1981.
Jackson, M. J., and L. Tauxe, Anisotropy of magnetic susceptibility and remanence: Developments in the characterization of tectonic, sedimentary, and igneous fabric, *Rev. Geophys.*, *29*, 371–376, suppl., 1991.
Jackson, M., G. J. Borradaile, P. J. Hudleston, and S. K. Banerjee, Experimental deformation of synthetic magnetite-bearing calcite sandstones: Effects on remanence, bulk magnetic properties and magnetic anisotropy, *J. Geophys. Res.*, *98*, 383–401, 1993.
Jelinek, V., Statistical processing of anisotropy of magnetic susceptibility measured on groups of specimens, *Stud. Geophys. Geod.*, *22*, 50–62, 1978.
Jelinek, V., Characterization of the magnetic fabrics of rocks, *Tectonophysics*, *79*, T63–T67, 1981.
King, J. G., and W. Williams, Low-temperature magnetic properties of magnetite, *J. Geophys. Res.*, *105*, 16,427–16,436, 2000.
Kirschvink, J. L., The least squares line and plane and the analysis of paleomagnetic data, *Geophys. J. R. Astron. Soc.*, *62*, 699–718, 1980.
Kodama, K. P., Remanence rotation due to rock strain and the stepwise application of the fold test, *J. Geophys. Res.*, *93*, 3357–3371, 1988.
Krogh, T. E., and D. E. Moser, U-Pb zircon and monazite ages from the Kapuskasing uplift: Age constraints on deformation within the Ivanhoe Lake fault zone, *Can. J. Earth Sci.*, *31*, 1096–1103, 1994.
Lowrie, W., A. M. Hirt, and R. Kligfield, Effects of tectonic deformation on the remanent magnetisation of rocks, *Tectonics*, *5*, 713–722, 1986.
MacDonald, W. D., Net tectonic rotation, apparent tectonic rotation, and the structural tilt correction in paleomagnetic studies, *J. Geophys. Res.*, *85*, 3659–3669, 1980.
McWhinnie, S. T., B. A. van der Pluijm, and R. van der Voo, Remagnetisations and thrusting in the Idaho-Wyoming overthrust belt, *J. Geophys. Res.*, *95*, 4551–4559, 1990.
Merrill, R. T., and M. W. McElhinny, *The Earth's Magnetic Field*, 401 pp., Academic, San Diego, Calif., 1983.
Mitchell, R. H., The lamprophyre facies, *Mineral. Petrol.*, *51*, 137–146, 1994.
Morrison, G. W., Characteristics and tectonic setting of the shoshonite rock association, *Lithos*, *13*, 97–108, 1980.
Muxworthy, A. R., and E. McClelland, The causes of low-temperature demagnetization of remanence in multidomain magnetite, *Geophys. J. Int.*, *140*, 115–131, 2000.
Stamatakos, J., and K. P. Kodama, The effects of grain-scale deformation on the Bloomsburg formation pole, *J. Geophys. Res.*, *96*, 17,919–17,933, 1991.
Symons, D. T. A., and T. A. Vandall, Paleomagnetic evidence for Proterozoic tectonism in the Kapuskasing Structural Zone, Ontario, *J. Geophys. Res.*, *95*, 19,199–19,211, 1990.
Symons, D. T. A., M. T. Lewchuk, D. J. Dunlop, V. Costanzo-Alverez, H. C. Halls, M. P. Bates, H. C. Palmer, and T. A. Vandall, Synopsis of paleomagnetic studies in the Kapuskasing structural zone, *Can. J. Earth Sci.*, *31*, 1206–1217, 1994.
Tarling, D. H., and F. Hrouda, *The Magnetic Anisotropy of Rocks*, 217 pp., Chapman and Hall, New York, 1993.
Vandall, T. A., and D. T. A. Symons, Paleomagnetism of Archean granites and Matachewan dikes in the Wawa subprovince, Ontario: Re-evaluation of the Archean apparent polar wander path, *Can. J. Earth Sci.*, *27*, 1031–1039, 1990.
van der Voo, R., *Paleomagnetism of the Atlantic, Tethys and Iapetus Oceans*, 411 pp., Cambridge Univ. Press, New York, 1993.
Werner, T., and G. J. Borradaile, Paleoremanence Dispersal across a transpressed Archean Terrane: Deflection by anisotropy or by late compression?, *J. Geophys. Res.*, *101*, 5531–5545, 1996.
Williams, H. R., The Quetico subprovince, in *The Geology of Ontario*, *Spec. Publ.4*(2), pp. 383–403, Ont. Geol. Surv., Sudbury, 1992a.
Williams, H. R., The Wawa subprovince, in *The Geology of Ontario*, *Spec. Publ.4*(2), pp. 485–539, Ont. Geol. Surv., Sudbury, 1992b.
Williams, H. R., G. M. Scott, P. C. Thurston, R. H. Sutcliffe, G. Bennett, R. M. Easton, and D. K. Armstrong, Tectonic evolution of Ontario: Summary and synthesis, in *The Geology of Ontario*, *Spec. Publ.4*(2), pp. 1255–1334, Ont. Geol. Surv., Sudbury, 1992.
Woodcock, N. H., Specification of fabric shapes using an eigenvalue method, *Geol. Soc. Am. Bull.*, *88*, 1231–1236, 1977.
Wyman, D. A., and R. Kerrich, Archean lamprophyres of the Superior Province, Canada: Distribution, petrology, and geochemical characteristics, *J. Geophys. Res.*, *94*, 4667–4696, 1988.

G. J. Borradaile and T. J. Lemmetty, Geology and Physics Department, Lakehead University, Thunder Bay, Ontario, Canada P7B 5E1. (borradaile@lakeheadu.ca)

T. Werner, Institute of Geophysics, Polish Academy of Sciences, 01452 Warsaw, Poland. (twerner@igf.edu.pl)

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